1 THE FLOOR THEOREM

1.1 A(q)   P(q) unconditionally: semicontinuity of integer rank, plus the Point-Count identity

The Sofa Theorem — standalone development of Proposition 1.3 · 9 July 2026 Everything here is either proved on the page, verified byte-exact against archived data (named), or explicitly imported from [DS, arXiv:1405.4683] with the import stated.

1.1.1 1. Statement

Let `S = F [s ..s ]`, `E = (e ,e ,e )`, `q = 3^v`, and `A(q) = dim_{F } S/(E + m^{[q]})`. Let `P(q) = 15q³-45q²+55q−24`. > **Theorem (Floor).** `A(q)   P(q)` for every `q = 3^v`.

1.1.2 2. The three ingredients

(I) A(q) as a co-rank of an INTEGER matrix. `dim_{F } S/m^{[q]} = q` (monomials with exponents $<$ q). Inside this truncation, `A(q) = q  - rank_{F } G_q`, where `G_q` is the matrix whose columns are the truncated multiples `{ s^ ·e_k mod m^{[q]} : k   {1,3,5}, | | arbitrary }` expressed in the monomial basis. The entries of `G_q` are the integer coefficients of the elementary symmetric polynomials and their monomial shifts — **`G_q` is defined over** (entries 0 and 1), and its reduction mod 3 computes A(q). This is definitional.

(II) Rank can only DROP under reduction mod p. For any integer matrix, `rank_{F }(G mod 3)   rank_ (G)`: a non-vanishing minor over F lifts to a minor with determinant   0 (mod 3), hence   0 over   . Therefore > `A(q) = q  - rank_{F }(G_q)   q  - rank_ (G_q) = dim_  S_ /(E + m^{[q]})`. The right side is the SAME quotient computed in characteristic 0. (Semicontinuity: special fibers can only get bigger.)

(III) The characteristic-0 dimension IS P(q). This is the [DS] framework's target: `P` is the Degtyarev–Shimada rank polynomial, the characteristic-0 count for the degree-3^v Fermat setting — the quantity their Conjecture 1.2 asserts is preserved in characteristic 3. We do not re-derive their characteristic-0 computation; we import it as the definition of the target, and we CORROBORATE it independently two ways: - **(III-a) The Point-Count identity (proved and sealed in this project, all v):** `P(3^v) = |X_v(F )|`, where `X_v =  _J (L_J)^v` is the union of the 15 matching-subspace powers. Proof: exact inclusion–exclusion over finite sets + the Frozen-Lattice intersection identity `| _{J T}(L_J)^v(F )| = 3^{v·c(T)}` (c(T) = bipartite-component count of the edge-union graph — a graph invariant, frozen across the tower). Grouping by c(T) exhibits `|X_v(F )|` as a cubic polynomial in `3^v` with coefficients `(15, -45, 55, -24)`, computed by exhaustive enumeration of all 2^1 −1 subsets. Verified byte-exact: `|X_v(F )| = 141, 7761, 263901 = P(3), P(9), P(27)` at v = 1,2,3 (engine `V32_GATE_P_pointcount.py`, archived with logs). - **(III-b) The v=1 rational rank, byte-exact:** the archived assembly matrix (`DeltaZ_v1_data.py`, 405×729, md5 `ace040f7296b7f323ccc81c8cdf34f0a`) has exact rational rank computed with `fractions.Fraction` arithmetic consistent with `P(3) = 141`, matching the mod-3 computation `A(3) = 141` — i.e., at v = 1 the floor is attained with equality, as the theorem's conclusion requires.

1.1.3 3. Proof of the Floor

Combine (I) + (II): `A(q)   dim_  S_ /(E + m^{[q]})`. By (III) the right side is `P(q)`.

1.1.4 4. What is proved here vs. what is imported — stated exactly

- Proved on this page, v: (I) the integer-matrix presentation; (II) the rank semicontinuity; (III-a) the Point-Count identity `P(3^v) = |X_v(F )|` with its inclusion–exclusion proof and three byte-exact checkpoints.
- Imported from [DS]: that `P` is the characteristic-0 dimension of the target (their rank polynomial — the very quantity Conjecture 1.2 is about). A referee should confirm this identification against arXiv:1405.4683 §1; it is the definition of the problem, not a step of our proof.
- Independence: nothing in this floor uses Sections 2–5 of the main paper (Recognition, duality, collar, Ledger). No circularity is possible: the floor is older than, and disjoint from, the ceiling machinery.

Files: V32_GATE_P_pointcount.py (+log), DeltaZ_v1_data.py (md5 above), THE_POINT_COUNT_THEOR THE_FROZEN_LATTICE_THEOREM_standalone.md — all in the repository.